Yandex

Yandex

Yandex Translate

Natural Language Processing

David Talbot

ShAD 2019

Introduction

- What is NLP?

Introduction

- What is NLP?
- Why is it hard?

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- What is NLP?
- Why is it hard?
- Some of the Story So Far

Natural Language Processing

Natural Language Processing

- Machine Learning

Natural Language Processing

- Machine Learning
- Engineering

Natural Language Processing

- Machine Learning
- Engineering
- Computational Linguistics

Why NLP is Hard?

- Ambiguity

Яндекс Переводчик

ТЕКСТ САЙТ КАРТИНКА

АНГЛИЙСКИЙ

Finally a computer that understands you like your mother.

57 / 10000

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Наконец-то компьютер, который понимает, что ты любишь свою мать.

Перевести в Google Bing

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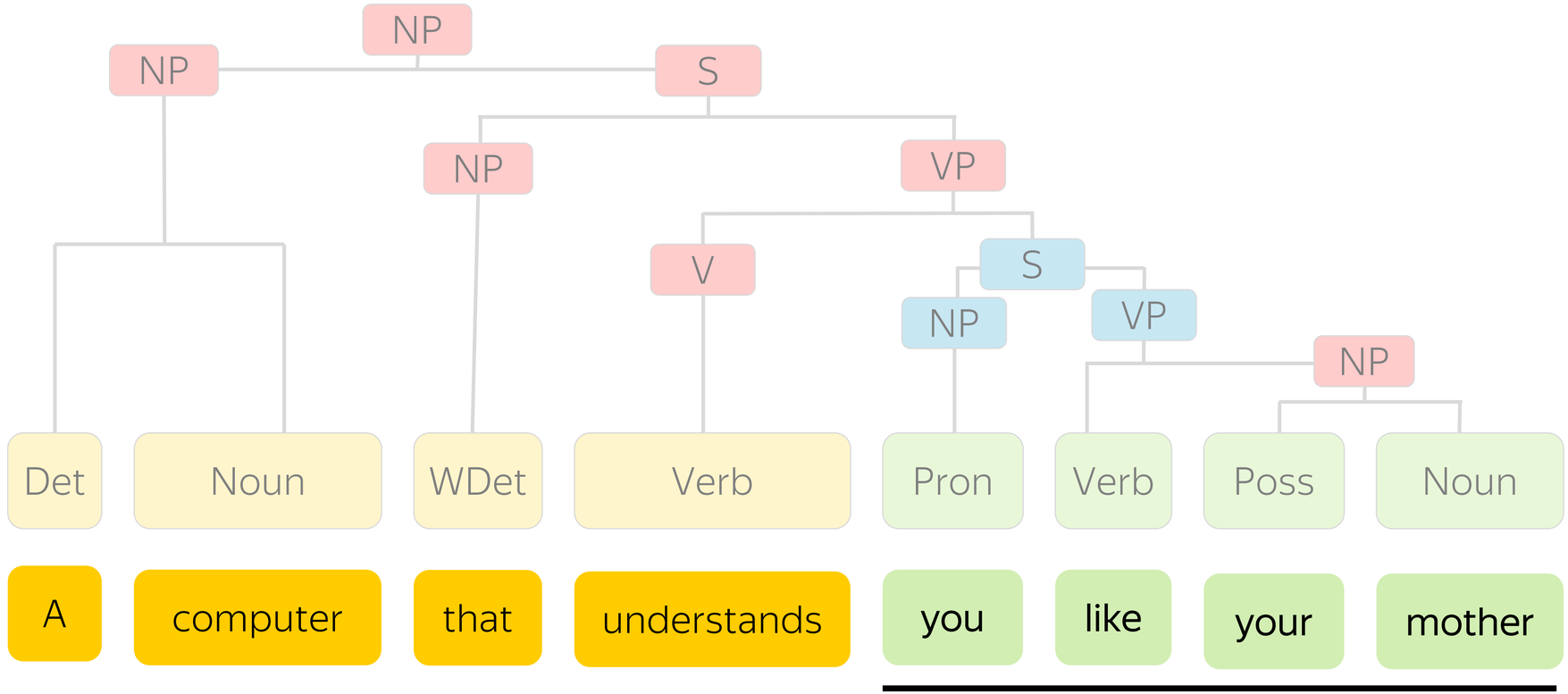
Наконец-то компьютер, который понимает тебя, как твоя мать.

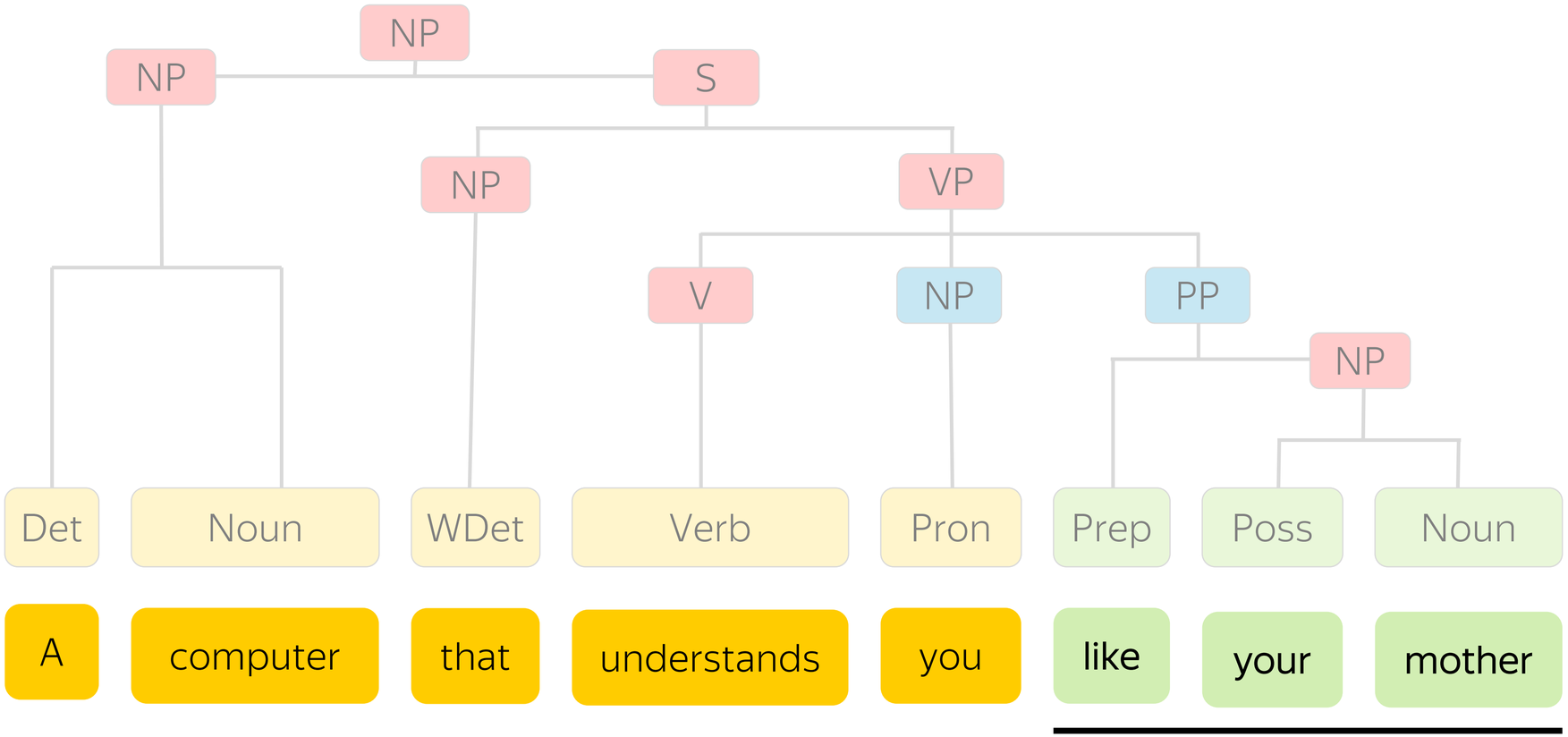
Перевести в Google Bing

Ambiguity



Компьютер, который понимает, что ты любишь свою мать.

Ambiguity



Компьютер, который понимает тебя так же, как твоя мама.

Why NLP is Hard?

- Ambiguity
- Structure

Яндекс Переводчик

ТЕКСТ САЙТ КАРТИНКА

АНГЛИЙСКИЙ

The animal didn't cross the street because it was too tired.

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Яндекс Переводчик

ТЕКСТ САЙТ КАРТИНКА

АНГЛИЙСКИЙ

The animal didn't cross the street because it was too tired.

РУССКИЙ

Животное не переходило улицу, потому что слишком устало.

Перевести в Google Bing

Яндекс Переводчик

ТЕКСТ САЙТ КАРТИНКА

АНГЛИЙСКИЙ

The animal didn't cross the street because it was too wide.

59 / 10000

Яндекс Переводчик

ТЕКСТ САЙТ КАРТИНКА

АНГЛИЙСКИЙ

The animal didn't cross the street because it was too wide.

РУССКИЙ

Животное не переходило улицу, потому что она была слишком широкой.

Перевести в Google Bing

Why NLP is Hard?

- Ambiguity
- Structure
- Linguistic diversity

Syntactic Differences

“Early must I rise. Leave now you must!”
— Yoda

Syntactic Differences

“Early must I rise. Leave now you must!”
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私は早起きする必要があります。 今すぐ出て行け!

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I [topic] early rise must . [You] now leave must.

Syntactic Differences

“Early must I rise. Leave now you must!”
— Yoda

私は早起きする必要があります。 今すぐ出て行け!

I [topic] early rise must . [You] now leave must.

I must rise early. You must leave now.

Story so far...

- Information Theory (1940s)

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- Distributional Semantics (1940s)

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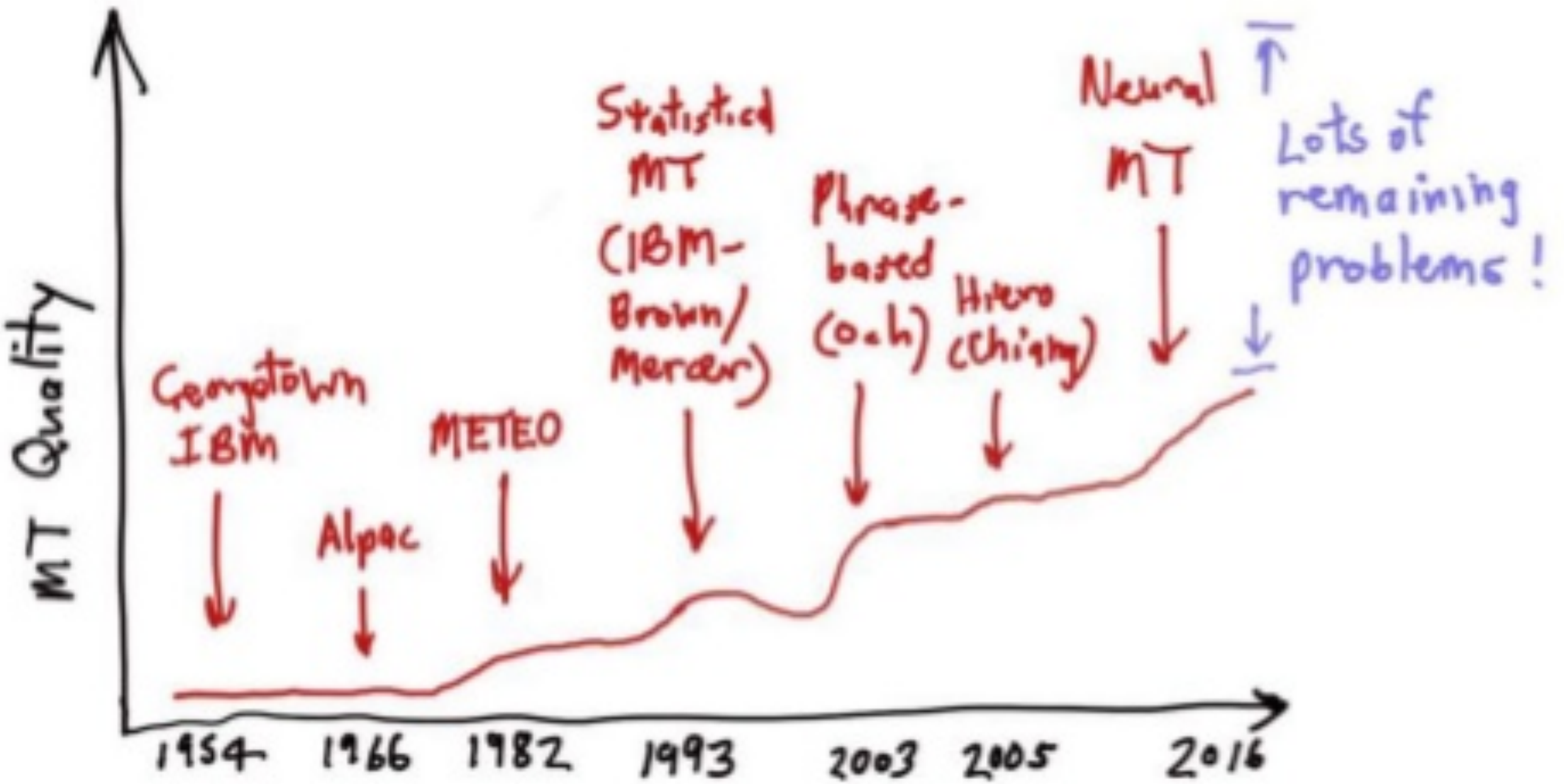
Story so far...

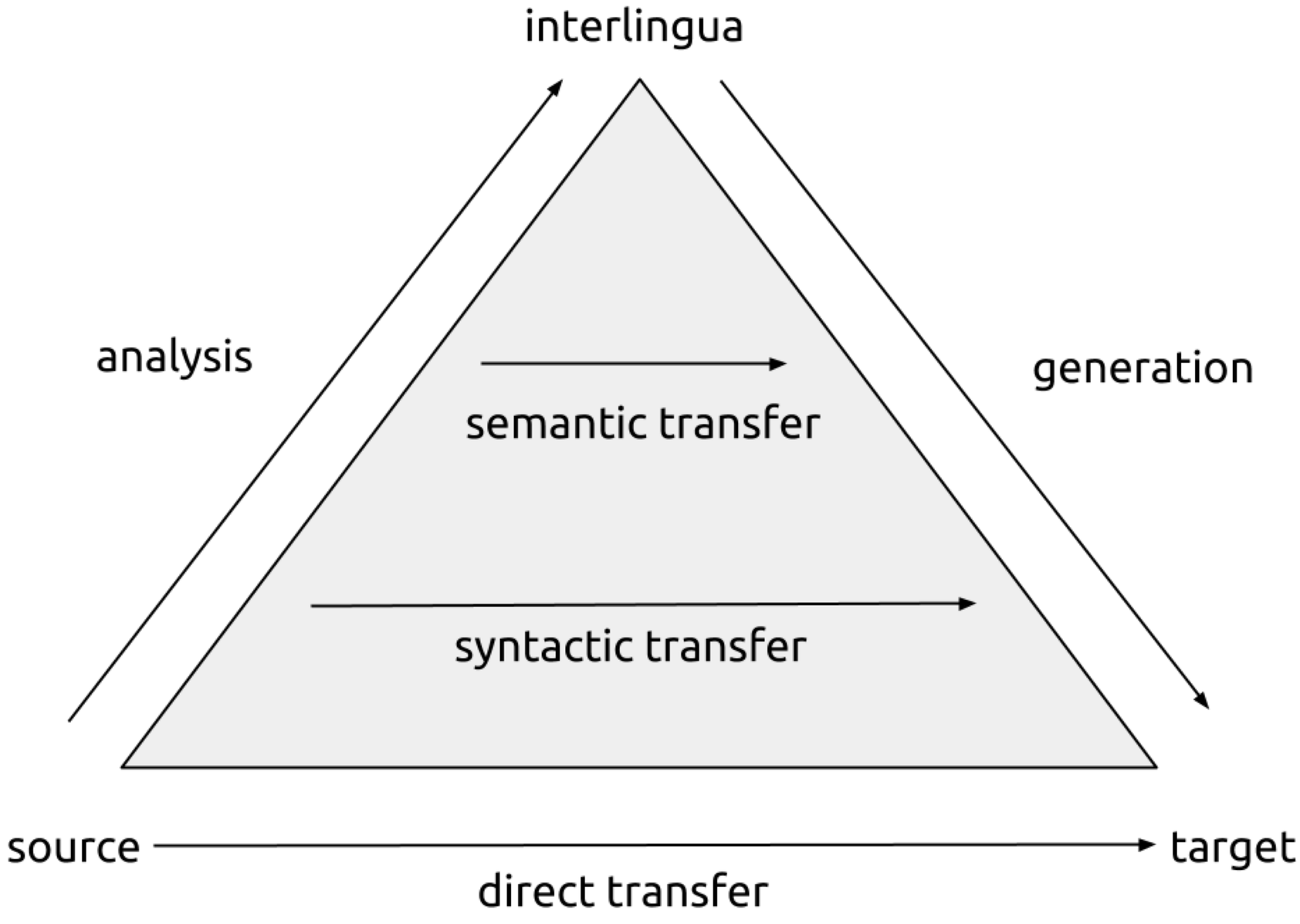
- Information Theory (1940s)
- Distributional Semantics (1940s)
- Rule-based Machine Translation (1950s)
- Rule-based AI (1960s-1970s)
- Statistical Speech Recognition (1980s)
- Statistical Machine Translation (1990s)

Progress in MT

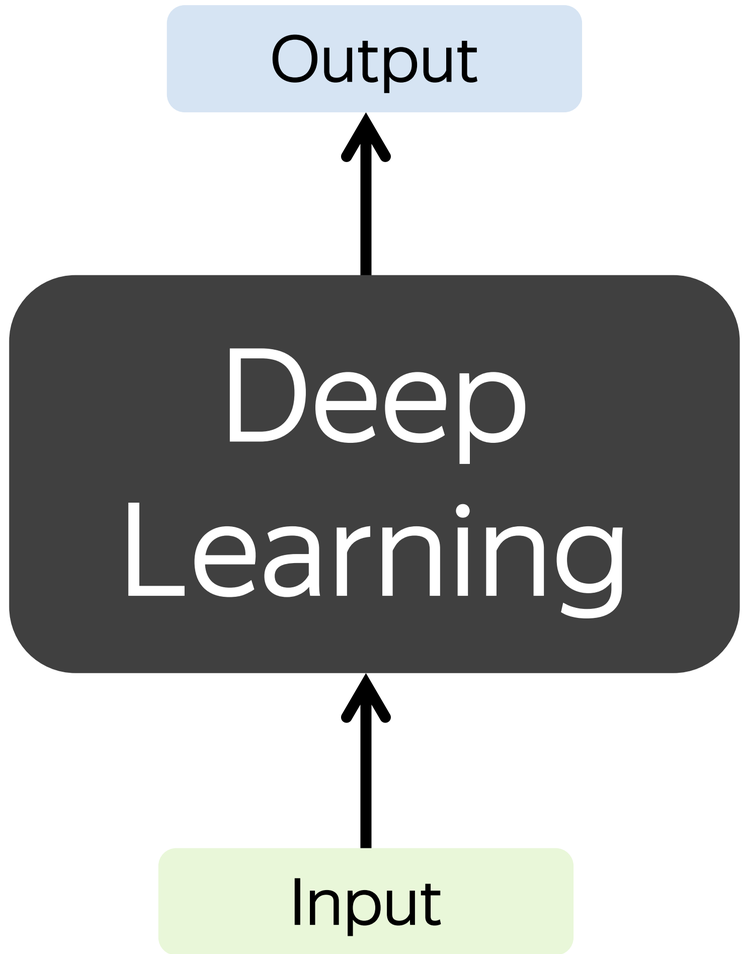


Credit: Chris Manning 2016

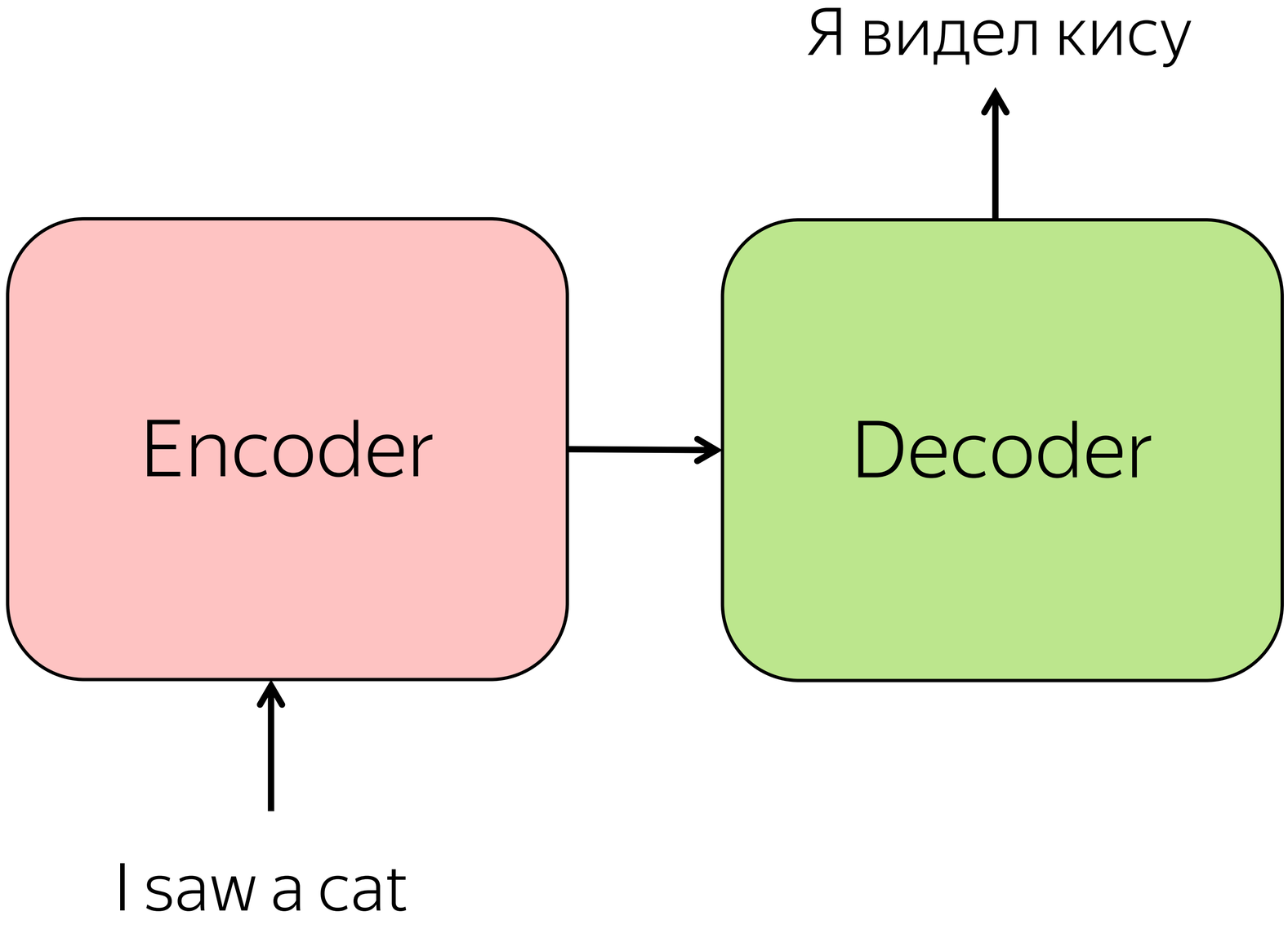
Vauquois Triangle



Deep Learning and NLP



Encoder Decoder



Deep Learning and NLP

- Representation Learning (e.g. word embeddings)

Deep Learning and NLP

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Deep Learning and NLP

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- End-to-end Optimization (e.g. NMT)
- Transfer Learning (e.g. BERT)
- Structure Learning (e.g. Transformer)